candidates to have hands-on expertise with React and to demonstrate proficiency in building complex, production-ready applications.

As we go, we will examine the core concepts and ecosystem of React, always with an eye towards the important points that need to be addressed in an interview. This will help candidates use these features effectively and discuss them eloquently, demonstrating a comprehensive understanding of React's inner workings.

Several factors contribute to React's ongoing success (https://survey.stackoverflow.co/2024/technology#2-web-frameworks-and-technologies):

- *Popularity*—React remains the most desired framework, with 33.4% of developers wanting to learn and use it more, surpassing Angular (13.9%) and Vue (16.3%).
- *Performance*—React's Virtual DOM and efficient rendering mechanism contribute to its high performance, particularly for complex and dynamic applications.
- *Flexibility*—React's component-based architecture and unidirectional data flow provide developers with a flexible and maintainable approach to building user interfaces.
- *Community and ecosystem*—React boasts a large and active community, as well as an extensive ecosystem of third-party libraries, making it a versatile choice for developers.
- *TypeScript support and tooling*—React offers TypeScript support and integrates seamlessly with modern tooling, including ESLint, Prettier, and advanced build tools like Vite and Webpack.
- *Industry adoption*—Major companies, such as Facebook, Twitter, and Instagram, use React, further solidifying their position as an industry standard.
- *Learning curve*—While React has a learning curve, it is generally considered manageable compared to other frameworks, such as Angular.
- *Job market demand*—React continues to be the most sought-after JavaScript technology in 2025, indicating strong demand for React developers in the job market.

Figure 1.2, based on npm trends data, highlights React's clear dominance in